EXPERIMENT 5

Implement and compare the performance of three popular CNN architectures: VGG, ResNet, and GoogLeNet for classification using Dogs vs. Cats dataset. Suggested Dataset: Dogs vs. Cats dataset.

```
import tensorflow as tf
from tensorflow.keras import layers, models
from tensorflow.keras.applications import VGG16, ResNet50

# Load CIFAR-10 dataset
(x_train, y_train), (x_test, y_test) = tf.keras.datasets.cifar10.load_data()

# Use smaller data for phone
x_train = x_train[:10000] / 255.0
y_train = y_train[:10000]

x_test = x_test[:2000] / 255.0
y_test = y_test[:2000]

# Simple CNN
cnn = models.Sequential([
    layers.Conv2D(32, (3,3), activation='relu',
                  input_shape=(32,32,3)),
    layers.MaxPooling2D(2,2),

    layers.Conv2D(64, (3,3), activation='relu'),
    layers.MaxPooling2D(2,2),

    layers.Flatten(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

cnn.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

print("Training Simple CNN")
cnn.fit(x_train, y_train, epochs=2, batch_size=32)

loss, acc = cnn.evaluate(x_test, y_test, verbose=0)
print("CNN Accuracy:", acc)


# VGG16
vgg = VGG16(
    weights='imagenet',
    include_top=False,
    input_shape=(32,32,3)
)

vgg.trainable = False

vgg_model = models.Sequential([
    vgg,
    layers.GlobalAveragePooling2D(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

vgg_model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

print("\nVGG16 Model Created")


# ResNet50
resnet = ResNet50(
    weights='imagenet',
    include_top=False,
    input_shape=(32,32,3)
)
```

```
resnet.trainable = False

resnet_model = models.Sequential([
    resnet,
    layers.GlobalAveragePooling2D(),
    layers.Dense(64, activation='relu'),
    layers.Dense(10, activation='softmax')
])

resnet_model.compile(
    optimizer='adam',
    loss='sparse_categorical_crossentropy',
    metrics=['accuracy']
)

print("ResNet50 Model Created")

print("\n----- MODEL COMPARISON -----")
print("Simple CNN Accuracy :", acc)
print("VGG16 and ResNet50 models created successfully")
```

```
Training Simple CNN
Epoch 1/2
313/313 ━━━━━━━━━━━━━━━━━━━━ 18s 49ms/step - accuracy: 0.3185 - loss: 1.8384
Epoch 2/2
313/313 ━━━━━━━━━━━━━━━━━━━━ 18s 40ms/step - accuracy: 0.4754 - loss: 1.4531
CNN Accuracy: 0.45399999618530273

VGG16 Model Created
ResNet50 Model Created

----- MODEL COMPARISON -----
Simple CNN Accuracy : 0.45399999618530273
VGG16 and ResNet50 models created successfully
```